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Guiding supervisors in artificial intelligence-enabled forecasting: Understanding the impacts of salience and detail on decision-making☆

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ABSTRACT

In many real-world situations, multiple humans are involved in decision-making when interacting with machine recommendations. We investigated a setting where an artificial intelligence system creates demand forecasts that a human planner can either accept or revise, and a supervisor then makes the final decision about which forecast to select. We designed and conducted two experimental studies to understand decision-making by a supervisor. First, we provided the improvement probabilities of adjustments at an aggregated level and found evidence for overoptimism bias and mean anchoring. Second, we provided decomposed guidance based on two adjustment attributes, direction and magnitude, to investigate the role of salience based on the distance between the improvement probabilities and level of detail in guidance effectiveness. We found no significant difference in using less and more salient guidance provided that the detail level was fixed. However, revealing more details when the guidance was more salient increased the use of guidance.

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1. Introduction

Advances in computing power and the maturity of machine-learning approaches have promoted the widespread use of artificial intelligence (AI) in many applications, including in medical diagnostics (Jussupow, Spohrer, Heinzl, & Gawlitza, 2021), process quality improvement (Senoner, Netland, & Feuerriegel, 2021), empowering digital platforms (Rai, Constantinides, & Sarker, 2019), and retail demand forecasting (Cui, Gallino, Moreno, & Zhang, 2018). The quality of the recommendations obtained from AI has steadily improved over time, but the final decision is still often made by a human expert who is ultimately held responsible for the outcome (Alicke, Hoberg, & Rachor, 2019). Therefore, humans are typically free to adjust and override the decisions proposed by machines. Previous research into human–machine interactions has shown that the fraction of decisions altered by humans varies widely among application areas and it is influenced by factors such as the decision quality of a system, the perceived risk of relying on decisions made by a system, or the specific context (De Baets & Harvey, 2020; Fildes, Goodwin, Lawrence, & Nikolopoulos, 2009; Goodwin, Gönül, & Önkal, 2013). Due to the importance of human judgment in forecasting, it is crucial to understand when adjustments to system-generated forecasts actually contribute to the higher accuracy of forecasts.

When interacting with machine recommendations, multiple stakeholders are often directly involved in the

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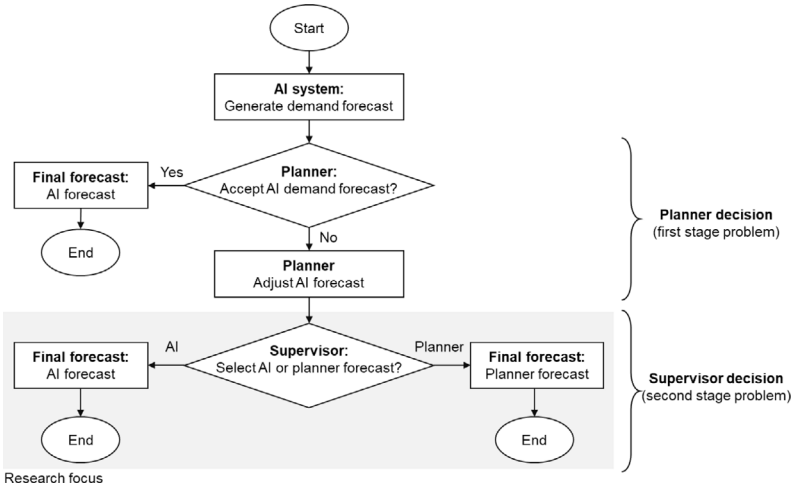


Fig. 1. Flow chart illustrating decisions in a research setting.

decision (Önkal, Goodwin, Thomson, Gönül, & Pollock, 2009; Perera, Aruchunarasa, & Önkal, 2022; Perera, Hurley, Fahimnia, & Reisi, 2019; van Dijk & Franses, 2019). For instance, in healthcare diagnostics, AI algorithms such as Google's DeepMind can predict diseases by analyzing medical images or data. A doctor can also make a diagnosis based on his or her expertise. When these results conflict, a second doctor or a medical review board may then decide whether to follow the AI's diagnosis or the first doctor's diagnosis (Powles & Hodson, 2017). In the context of recruitment and hiring, some companies use AI algorithms to screen résumés and predict the success of job candidates. A hiring manager can also evaluate résumés based on his or her professional judgment. A human resources director should then audit the decision to avoid biases (Dastin, 2018; Rhea et al., 2022). In these cases, a final decision-maker would consider the strengths and weaknesses of earlier predictions made by both AI and humans. AI can process vast amounts of data quickly and identify patterns that humans might miss, but it can also cause errors or overlook nuances that humans would notice. However, human judgment can be influenced by bias or error, but it can also incorporate context and nuance in ways that are currently possible for AI (Rastogi, Zhang, Wei, Varshney, Dhurandhar, & Tomsett, 2022).

In the present study, similar to these examples, we considered a setting with two decision-makers in the context of retail demand forecasting. Fig. 1 shows the flow of decisions. Initially, an AI system creates the demand forecast for an SKU. The first human decision-maker, a *planner*, decides whether to keep the forecast provided by an AI system or to override it. No further action is needed if the planner chooses to keep the AI forecast. However, if the planner decides to deviate from the AI forecast, a second human decision-maker, a *supervisor*, is presented with both forecasts. The supervisor makes the final decision about which forecast to select: the original AI-generated forecast or the human-adjusted forecast (see Fig. 2 for an example). In contrast to the planner, the supervisor making the final decision can be somewhat detached from the operational considerations of the front line. Therefore, guidance regarding the past performance of adjustments can support the supervisor's decision-making process. Guidance regarding the quality of overrides has considerable potential to improve the quality of final decisions, but few previous studies have investigated this issue.

In our study, we focused on the supervisor's decision-making process and tested alternative types of guidance. In the first-stage problem for the planner, we used real-world decisions made by the human planners of leading European retailers who could override forecasts from an advanced AI system (Khosrowabadi, Hoberg, & Imdahl, 2022). In the original setting, planners overrode 5.1% of the AI-generated forecasts, and approximately 49.9% of the adjustments were valuable. Further analysis highlighted that the expected value of an override varied widely based on the attributes of the adjustment, especially its direction (downward vs. upward) and magnitude (small vs. large). Therefore, guiding supervisors in the second-stage problem about when adjustments are valuable seems critical.

We conducted two experiments to understand the second-stage problem with decision-making by the supervisor and the supervisor's reaction to guidance. We designed guidance by communicating the improvement probability of adjustments, i.e., the likelihood of improved accuracy when a forecast is adjusted. In Study 1, we analyzed the subjects' natural behavior in deciding between an AI forecast and a human-adjusted forecast when guidance was provided at an aggregated level. In this neutral setting, the value of an adjustment is equal to that of no adjustment, i.e., the improvement probability of an adjustment is 50%. Thus, we aimed to understand whether subjects naturally preferred to select a human-adjusted forecast for specific attributes. In Study 2, we aimed to understand how the effectiveness of guidance depends on the salience and detail of the guidance. Therefore, we decomposed the guidance based on adjustment attributes. We provided subjects with the improvement probabilities of adjustments, which differed according to direction (upward vs. downward) and magnitude (small

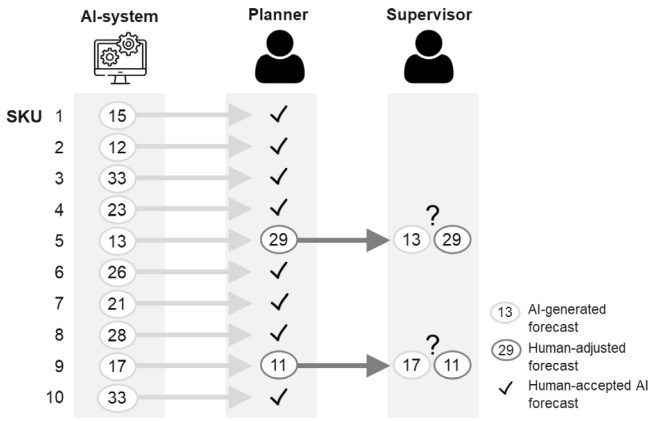


Fig. 2. Examples of choices in the research setting.

vs. large). To vary the salience, we changed the improvement probabilities provided. In a more salient condition, we provided a more significant difference in the improvement probabilities between upward and downward adjustments. Thus, we manipulated how noticeable the guidance was to the supervisor. We also varied the level of detail in the guidance. Guidance could have improvement probabilities based solely on the direction (upward or downward), which was less detailed because it did not distinguish between large and small adjustments. Alternatively, guidance could include both the direction and magnitude (upward large, upward small, downward large, or downward small) to convey more detailed information. More efficient guidance should support supervisors. Accordingly, the efficiency of guidance entails a tradeoff between the time that the decision-maker requires to process the guidance and the contribution of the guidance to the quality of the decision. Therefore, the information communicated should be concise yet effective (Qiu, Pang, & Lim, 2012; Sroginis, Fildes, & Kourentzes, 2019). Sharing an excessively large amount of information can affect the guidance efficiency and cause cognitive overload. By contrast, an excessively small amount of information can detrimentally affect decision-making because of the lack of essential details. We examined how the effectiveness of guidance depended on the complexity of the information provided and whether it was moderated by the salience of the guidance.

We obtained three main findings. First, we found that when the improvement probability was provided at an aggregated level, subjects manifested overoptimism because they naturally selected more human-adjusted forecasts when adjustments were made in the upward direction. Second, we observed that subjects tended to choose forecasts closer to the mean demand, which is consistent with the well-known mean-anchoring effect (Bloomfield & Kulp, 2013; Schweitzer & Cachon, 2000). Third, and most importantly, when dealing with decomposed guidance, we found that revealing more details when the guidance was more salient increased its use. However, giving less salient improvement probabilities did not significantly increase the use of guidance. Therefore, revealing more information only helped when subjects could clearly distinguish among alternative decisions. Providing more detailed information may require additional effort, so this finding clearly shows how more efficient and effective guidance can be designed.

Our research extends current understanding by introducing a novel experimental design that considers scenarios involving human collaborative interactions with algorithmic recommendations. Despite their significant practical implications, these scenarios have not been studied extensively. Building on the foundational work of Dietvorst, Simmons, and Massey (2015) and Perera et al. (2022), who investigated human responses to advice from machines versus responses to other humans, our experiment focused on the development of strategic guidance for supervisors tasked with choosing between AI-generated and human-generated forecasts. We manipulated two key aspects of this guidance, the salience and level of detail, to methodically assess their impacts on the decisions of forecast supervisors. Our findings augment the existing academic discourse as well as offering tangible recommendations for developing guidance frameworks in practice.

The remainder of this paper is organized as follows. In Section 2, we review relevant literature to identify factors that should be considered when formulating guidance, and we formulate our hypotheses. In Section 3, we explain the experimental design and results of Study 1, where neutral guidance about improvement probability was provided at an aggregated level. Section 4 presents Study 2, which extended Study 1 and tested the effectiveness of guidance about the improvement probability combined with the direction and magnitude of the adjustments. In Section 5, we give our conclusions by highlighting the important findings and proposing directions for future research.

2. Literature review and hypothesis development

Two research streams are relevant to our study. First, we reviewed research into judgmental adjustments of forecasts to understand their context and when such adjustments are valuable. Second, we reviewed various types of decision guidance to better understand the effectiveness of giving guidance during decision-making.

2.1. Judgmental adjustments of forecasts

Human planners are held accountable for the accuracy of forecasts, so they often adjust system-recommended forecasts (Hoberg & Imdahl, 2022). Advanced systems already account for many driving factors (e.g., weather and planned promotions) when generating forecasts (Khosrowabadi et al., 2022), but human planners may have contextual information that cannot be easily considered in these systems, such as competitors' reactions, rumors, any statements with subjective belief (Sroginis, Fildes, & Kourentzes, 2023), as well as knowledge about market conditions, customers' behavior, or promotional effects. Influenced by optimism, pessimism, or risk aversion, planners can express their own preferences or opinions about future outcomes when making their adjustments Fildes, Goodwin, and De Baets (2023). The other reasons that Fildes et al. (2023) identified for human adjustments involve correcting perceived errors or biases in statistical forecasts, such as over- or underestimation, seasonality, or trends. Moreover, no historical data are available to feed into the system for a newly introduced product. Some events happen infrequently and it is not reasonable to intervene in the underlying system process to consider events such as celebrity support (Fildes, Goodwin, & Önkal, 2019). Furthermore, the black-box nature of AI systems may impact the degree to which a planner can trust a recommended forecast (Adadi & Berrada, 2018).

In addition to these factors related to systems and the variables that drive them, other factors are relevant to how humans interact with a system. Some humans are algorithm-averse (Allen & Choudhury, 2022; Dietvorst et al., 2015) whereas others are algorithm appreciators (Logg, Minson, & Moore, 2019). The former group may unfairly judge an error made by a system, overweight their own judgment (Dietvorst et al., 2015), or be eager to make adjustments due to overoptimism or overconfidence (Eroglu & Croxton, 2010; Franses & Legerstee, 2009; Ren & Croson, 2013). Optimism bias is a tendency to overestimate the likelihood of positive events and underestimate the likelihood of negative events (Kirshner & Shao, 2019). The latter group may rely on the computational power or accuracy of the system, and thus they have a higher degree of trust in using the recommended system forecasts. In this context, information about the performance of an algorithm and the possibility of holding control over decisions can affect the degree of algorithm aversion and algorithm appreciation. For example, the tendency to use an algorithm depends on how users are informed about the accuracy of the source of forecasts (Hou & Jung, 2021). Nevertheless, adjustments inevitably happen in practice.

Not all adjustments contribute to the forecast accuracy and some can actually degrade the accuracy. Previous research has identified a link between the performance of adjustments and the attributes of adjustments, particularly the direction and size of adjustments. In many cases, an increase in a system forecast (i.e., upward adjustment) performs worse than a decrease (i.e., downward adjustment) (Davydenko & Fildes, 2013; Fildes et al., 2009). This relationship becomes more evident with large adjustments because downward adjustments are bounded at zero, but upward adjustments have no upper bound. For instance, in the context of retail demand forecasting in a study by Khosrowabadi et al. (2022), among downward adjustments, the percentage of valuable adjustments with a large magnitude was 8.4% higher than those with a small magnitude; however, among upward adjustments, the percentage of valuable adjustments with a large magnitude was 26.8% lower than those with a small magnitude. Thus, considering these two attributes when giving supervisors guidance about the performance of adjustments is critical, and thus we incorporated these attributes into our experimental studies.

In the present study, we examined a setting where supervisors reviewed forecasts that were first generated by a system and later adjusted by human planners. Few other studies have investigated how subjects react when they encounter forecasts that other humans have already adjusted (Dietvorst et al., 2015; Önkal et al., 2009; Perera et al., 2022). Önkal et al. (2009) examined a similar setting to understand how providing subjects with a specific kind of forecast (unadjusted or adjusted together with explanations for any adjustments) affected the frequency and magnitude of further modifications by the subjects. They found that adjusted forecasts were less likely to be modified further and that adding explanations to the adjusted forecast could enhance their presumed reliability, and thus the acceptance of the forecasts without further modification. Our research differs from that conducted by Önkal et al. (2009) in two aspects. First, we did not give subjects an option to modify forecasts. Subjects could choose either an unadjusted (AI) forecast or an adjusted (human) forecast in each round. Second, in the study by Önkal et al. (2009), choosing human-adjusted forecasts was preferable to improve performance. However, we guided subjects toward choosing a better forecast by sharing different information based on the performance of adjustments in terms of their direction and magnitude. Accordingly, an adjusted forecast should not always be perceived as a better forecast unless indicated by the guidance.

Our study is similar to that conducted by Dietvorst et al. (2015) where the forecasting performance of both sources was shared to understand the reactions of subjects. They found that subjects mistakenly avoided using an algorithm when they knew it made mistakes, even when the algorithm actually outperformed its human counterpart. Based on a series of experiments, Perera et al. (2022) evaluated the effect of reporting the accuracy of decisions made by a system and the humans involved in adjusting those decisions. They considered a few layers of adjustments made by different human decision-makers to evaluate the role of multiple adjustments when information was provided about the accuracy of the decisions. Our study examined a similar setting but with a different goal. Instead of reporting a single number for the accuracy of the adjusted forecasts, we shared the aggregated performance of the adjustments in terms of how the improved probability depended on the attributes of adjustment because a single number may be insufficient to convey when an algorithm outperforms a human adjustment and vice versa; indeed, it may lead to an extreme solution of following one source of decision either most often or rarely. However, algorithms and humans have their own capabilities and should collaborate to yield the best result.

2.2. Decision guidance design

Decision guidance refers to any suggestion or information that aims to help subjects make better decisions in a particular task (Silver, 1991). Decision guidance can be suggestive, informative, or both, where suggestive decision guidance suggests an action, whereas informative decision guidance enlightens without suggesting an action (Bolton & Katok, 2008). According to a more general definition, in any ad hoc communication with subjects to improve their decisions, such as outcome feedback, the simplest form of the informative type can be a form of decision guidance. There are trade-offs between informative and suggestive types of guidance and how they can affect a user's perception and behavior (Goodwin, Fildes, Lawrence, & Stephens, 2011).

We formed our decision guidance as an informative type to direct a subject's attention toward choosing a better forecast. Salience and the details of guidance are two key factors that we varied in our decision guidance. It is important to carefully consider details, such as framing effects, to boost the performance of guidance and avoid unwanted effects (Arkes, Dawes, & Christensen, 1986; Levin, Schneider, & Gaeth, 1998). We did not aim to evaluate framing effects, so we only used a positive frame and remained consistent throughout our studies. To better understand the information that needs to be shared, it is essential to consider the inherent characteristics of the context where guidance will be used during its design, and there is no one-size-fits-all solution. In the forecasting context, uncertainty is a key inherent element that causes subjects to assess the risks of their different choices. A straightforward approach involves sharing the probability of success/failure or risk for each decision. For instance, Bolton and Katok (2018) tested the effectiveness of revealing the risk of taking an action in two formats: unequivocal (recommendations) and equivocal (probabilities). They found that they perform differently depending on the type of action suggested by the forecast guidance. The action can be no different from before as the status quo action, the best before getting the forecast as the preforecast optimal action, or different from before as the siren action,[1] but only if supported by the forecast for each period. The recommendations induce retaining the status quo more successfully, whereas probabilities induce the siren action more successfully. Recommendations of siren action are more vulnerable to the cry wolf effect where decision-makers lose trust in the guidance after repeated false alarms. Thus, sharing unequivocal recommendations can result in the cry wolf effect because of their higher perceived certainty. To avoid this effect, we designed our guidance by sharing the improvement probabilities of adjustments instead of recommending that subjects accept or decline one specific type of adjustment. In this manner, subjects would not feel deceived if a decision for a given forecasting period was not exactly compatible with the guidance because the probabilities were directly communicated to avoid triggering extreme interpretations of a recommendation's validity as never 0% or always 100%.

There is a general tendency to ignore advice, which is known as egocentric advice discounting. The likelihood of accepting guidance depends on the characteristics and preferences of the advice seekers and advice givers. Önkal et al. (2009) found that subjects tend to prefer advice from human experts rather than advice from statistical models, even when the latter are more accurate, possibly because of the perceived trustworthiness, credibility, and accountability of human advisors. Sniezek, Paese, and Switzer (1990) found that people tend to accept advice more when the advice giver is confident, consistent, and similar to them, which may be due to the perceived competence, reliability, and similarity of the advisors. In our experimental design, we referred to the system-generated forecast as the "AI forecast" and the adjusted forecast as the "Human forecast". We used these neutral terms in order to prevent subjects from inferring the characteristics of the advice giver. In addition, we emphasized the performance improvement comparison between the AI forecast and the Human forecast to encourage subjects to focus on performance outcomes and not on the source of advice.

2.3. Hypothesis development

To ensure best design guidance, we examined two factors that influenced its effectiveness in terms of the degree to which it was followed. The first of these factors was the value of information determined by the salience of the guidance. The second factor was the complexity of the information determined by the level of detail in the guidance. We varied the salience of the guidance by using the distance between improvement probabilities, and the level of detail in guidance by using the number of features provided in the guidance, direction, and magnitude of adjustments.

We first considered how *salience* affected guidance. To express salience, we provided different improvement probabilities for the direction and magnitude of adjustments. For example, we gave guidance in the following form: "The probability that the human forecast is better than the AI forecast differs for downward vs. upward adjustments. The improvement probabilities are about x% for downward adjustment and y% for upward adjustment". We expected that the discrepancy in the values of x% and y% would influence the salience of the information in the guidance.

When comparing different numbers from the same scale, a greater difference between the numbers makes it easier and faster for subjects to assess the difference by referring to the numerical distance effect (Moyer and Landauer, 1967; (Buckley & Gillman, 1974; Kaufmann et al., 2005; Pinel, Piazza, Le Bihan, & Dehaene, 2004)). For instance, subjects compared distant numbers such as five and nine faster and more accurately than close numbers such as five and six (Parkman, 1971). Therefore, a greater difference in the numbers can act as a stimulus to make the information from the numbers more salient.

[1] A decision or action that seems very appealing because it promises the best outcome according to a specific forecast or recommendation. However, this action is only optimal if the forecast is accurate.

Due to the numerical distance effect, when the difference between the probabilities is larger (vs. smaller), the information in the guidance becomes more salient. In addition, previous studies have confirmed the effect of salience bias, e.g., by enhancing adherence to price recommendations from a decision support system by shifting salience from inventory and sales to revenue (Caro & de Tejada Cuenca, 2023), thereby changing consumption by making taxes salient (Chetty, Looney, & Kroft, 2009), and improving the productivity of warehouse workers by making some KPIs salient (Weerasinghe, Perera, & Hurley, 2021). Thus, in accordance with the salience bias, when guidance is more conspicuous because of a greater discrepancy in probabilities, it has a stronger influence on the decisions of superiors according to their salience bias, and leads to higher utilization. Therefore, we considered the following hypothesis.

H1. *Providing more salient guidance will increase its usage compared with less salient guidance.*

Next, we considered the effect of the *detail* of guidance. In the experiment, we used two attributes: the direction and magnitude of adjustments. In a less detailed condition, we provided improvement probabilities only for adjustment direction (downward vs. upward). In a more detailed condition, improvement probabilities were given for both adjustment direction (downward vs. upward) and magnitude (small vs. large), where we presented improvement probabilities for four different combinations (downward small, downward large, upward small, and upward large).

Elaborating on the information in the guidance can increase transparency, which can then enhance the utilization of the guidance. In any decision-making task, a decision-maker needs to consider alternative courses of action, either through their own search or based on the information provided. Revealing more information is supposed to help decision-makers lessen the need to search. In a study by You, Yang, and Li (2022), which is similar to our present study, subjects were provided with both algorithmic and human forecasts. They appreciated the algorithm's advice more than the human's advice regardless of whether they knew the prediction performance at an aggregated level. However, the appreciation of the algorithm decreased after elaboration on the prediction performance of the advice source. In addition, an excessively small amount of information can be misleading because some potentially important aspects may be overlooked (Ungemach, Camilleri, Johnson, Larrick, & Weber, 2018). However, sharing further information that positively contributes to transparency could negatively increase the cognitive load of humans, i.e., decrease the available working memory, and thus reduce the degree of advice taking. Processing an excessively large amount of information is overwhelming for human decision-makers because of their limited cognitive ability (Kahneman, 2003). Information overload increases errors and negative effects such as confusion or frustration (Hunter, 2004; Jacoby, Speller, & Kohn, 1974). In this case, sharing more information helps but it can also result in an unwanted course of action, e.g., because of information bias as a human cognitive bias (Kleinbaum, Sullivan, & Barker, 2013). For example, when decision-makers are overwhelmed with an excessively large amount of information, they might focus on less relevant details, thereby causing them to make poor choices. This bias can manifest in various ways, such as misinterpreting data, overvaluing certain pieces of information, or ignoring critical factors. Consequently, instead of aiding the decision-making process, surplus information can cloud judgment and lead to actions that are not optimal or intended. We did not find any inconsistency between the degree to which people took an algorithm's advice over a human's advice provided that they were given the aggregated prediction performance, i.e., less detailed information (Castelo, Bos, & Lehmann, 2019; Dietvorst, Simmons, & Massey, 2018). However, an inconsistency arises in the case of further elaboration on performance: more detailed information (Dietvorst et al., 2015). In contrast to the study by You et al. (2022), Dietvorst et al. (2018) focused on factors that influenced the use of algorithms but they differed in terms of factors such as transparency and elaboration, algorithm performance, and results pertaining to satisfaction, confidence, trust, and delegation. You et al. (2022) found that elaboration on prediction performance decreased algorithm appreciation, which differed from Dietvorst et al. (2015) who focused on evaluating the effect of enabling participants to alter the algorithm's forecasts given its unelaborated prediction performance. The capacity for information handling by humans is limited and ranges from five to nine chunks of information when the size of a chunk depends on the type of information (Miller, 1956).

In summary, previous research has demonstrated the mixed effects of elaboration on decision performance. Although logically helpful, more information does not always enhance guidance effectiveness. Evidence suggests that more explicit information can improve efficacy, but more information is not always better. Inspired by Corman (1957), who suggested that providing more information that is less explicit may be as effective as less information that is more explicit, we investigated how the amount and salience of information affected guidance effectiveness. We predicted that more detailed guidance would increase usage when the information was more salient.

H2. *When guidance is more salient, providing more detailed guidance increases its usage.*

3. Study 1: Neutral and aggregated guidance

In Study 1, we provided guidance about the improvement probability at an aggregated level without any reference to the attributes of the adjustment made. We designed a neutral setting where the value of an adjustment was equal to that of no adjustment, i.e., 50% improvement probability for both human and AI forecasts. We aimed to understand the natural behavior of subjects when deciding between AI forecasts and human-adjusted forecasts, and whether they preferred a specific attribute of adjustment in their selection.

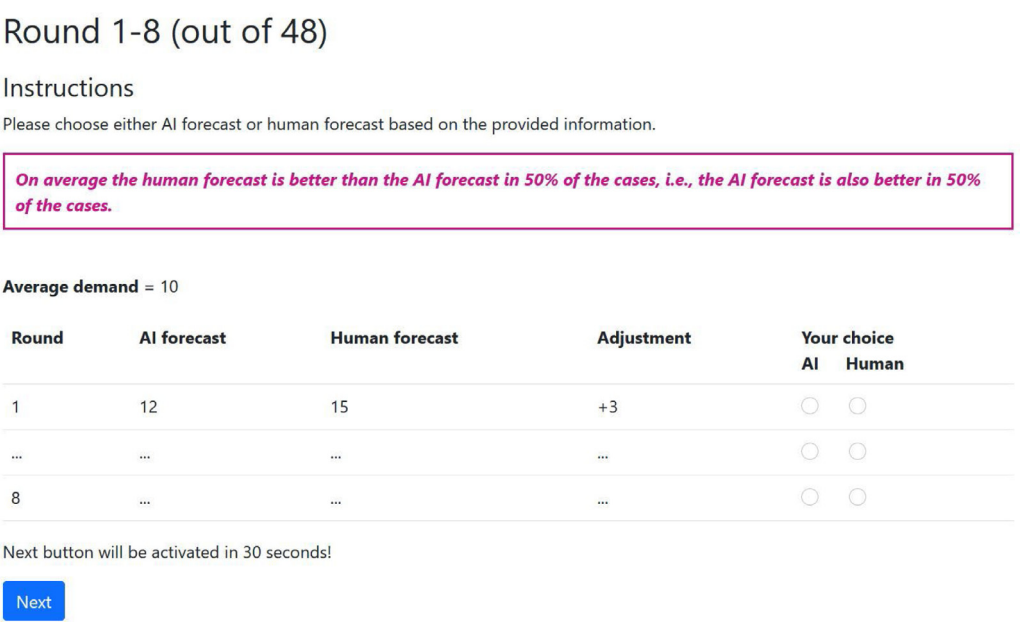


Fig. 3. User's screen for Study 1.

3.1. Experimental design

In our decision-making task, the subject took the role of a supervisor in a demand forecasting task. In each round, the subject received an AI forecast and a human-adjusted forecast, i.e., the forecast of the human planner was based on an adjustment to the AI forecast. In addition to these two forecasts, the following additional information was provided on the user's screen: (i) the average demand (which was kept constant at 10 units to reduce the task complexity) and (ii) the actual adjustment by the human calculated as the difference between the AI forecast and human forecast (i.e., human forecast – AI forecast). The adjustment was positive when humans adjusted the AI forecast upward and vice versa. Given this information, the subject's main task was to choose either an AI forecast or a human-adjusted forecast. Fig. 3 shows the user's screen for this study. In each round, the subject had the binary choice of using either the AI or the human-adjusted forecast.

In this study, the guidance stated the improvement probability of adjustment at an aggregated level. In particular, we provided a statement that read: "On average, the human forecast is better than the AI forecast in 50% of cases, i.e., the AI forecast is also better in 50% of cases". We accounted for framing effects, where we used a positive frame by alluding to improvement rather than deterioration and kept our statement neutral by mirroring the first sentence for the AI forecast. To increase the visibility of this guidance, we used a different text color and placed the statement in a box. The guidance was provided at the top of every page (see Fig. 3).

In the instructions for our studies, we included definitions of upward and downward as well as of small and large adjustments. Adjustments with an absolute magnitude of up to five were small, whereas adjustments with an absolute magnitude of more than five were large. Adjustments where the human forecast was higher than the AI forecasts were upward adjustments, whereas adjustments where the human forecast was lower than the AI forecasts were downward adjustments. For example, an adjustment of +2 was a small upward adjustment. We avoided stating medium adjustments for two reasons: considering them would increase the task complexity and require subjects to process more information; and medium adjustments were less critical than large adjustments, which had a larger impact, and small adjustments required inefficient expenditure of time and effort to achieve a minor effect. Appendix A contains all of the instructions provided for the study.

The study had six phases: (i) instructions and task descriptions to familiarize subjects with the task, process, and definitions of the variables involved; (ii) pre-experiment questions to check the subjects' understanding of the task and instructions (Appendix A); (iii) eight trial rounds so subjects could familiarize themselves with the task before the actual start; (iv) 48 main rounds; (v) post-experiment questions to check the consistency of subjects' choices and to measure their attitudes to AI (Appendix B lists these questions); and (vi) demographic questions.

As shown in Fig. 3, we asked subjects to finalize forecasts for eight rounds at a time in order to better manage their time across all 56 rounds, and this design was inspired by previous studies (Becker-Peth, Hoberg, & Protopappa-Sieke, 2020; Lurie & Swaminathan, 2009; Ren & Croson, 2013). To prevent subjects from rushing through the rounds, we set a 30-second waiting time for the appearance of the "next" button to proceed to the next page.[2] We also gave performance feedback every

[2] This was a minimum waiting time, and there was no time limit or pressure on subjects to force them to make a quick decision.

Table 1
Descriptive statistics.

Variable		Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	St. dev
AI forecast		1.0	8.0	12.0	11.8	14.0	26.0	5.6
Human forecast		2.0	7.8	11.5	11.9	15.3	26.0	6.0
Actual demand		1.0	7.0	10.0	10.5	13.0	22.0	4.7
Human Adjustment	Overall	−13.0	−4.5	0.0	0.1	4.5	11.0	6.3
	Upward Small	1.0	2.0	2.5	2.9	3.0	4.0	6.3
	Upward Large	6.0	6.5	8.5	8.2	10.5	11.0	6.3
	Downward Small[a]	1.0	2.0	2.5	3.0	3.0	4.0	6.2
	Downward Large[a]	6.0	7.0	8.0	7.8	9.0	13.0	6.3

[a] Absolute values are reported.

eight rounds to inform subjects of the results of their decisions and the calculated experimental coins earned per round and in total. The subjects were informed that they would be paid only after completing the entire task and submitting the completion code. We used two attention checks, with one at the beginning and another after the first 24 rounds. Early in the study, we filtered out inattentive subjects who incorrectly answered the first question used to check attentiveness. Those who answered correctly to the first check but not the second were allowed to continue and earn money; however, we excluded their inputs from our analysis. This design was validated in a small pilot study with 20 subjects.

In each round, subjects could earn coins by selecting the forecast closest to the actual demand. Earnings in experimental coins were calculated using the difference between the errors in both forecasts compared with the actual demand. For instance, if the AI forecast was eight, the human forecast 11, and the actual demand 13, a subject would earn coins only if the human forecast was selected and nothing if the AI forecast was selected. A subject who chose the human forecast would earn $|13 - -8| - |13 - -11| = 3$ coins. We used an exchange rate of 50 coins = USD 1 to calculate bonus payoffs in addition to a USD 1 participation fee. We coded the results using oTree (Chen, Schonger, & Wickens, 2016).

For the AI forecasts and human-adjusted forecasts, we used real-world data from a setting where an advanced AI system generated the initial forecasts that human planners could then adjust based on their experience or expertise (Khosrowabadi et al., 2022). In reality, only a fraction (5.1%) of all AI forecasts were adjusted; however, for our research purposes, we focused on those cases where AI forecasts were adjusted (also Figure see 2). Guidance performance was our main interest, so it was crucial to exclude potential confounds. Therefore, we chose a random sample from a real-world data set that complied with the requirements of the experiment (e.g., average demand, number of different adjustment types, and performance of human vs. AI forecasts). Please refer to Appendix C for the data sample.

Table 1 shows the basic statistics for our data sample from 48 main rounds. Subjects were presented with an equal number of different adjustments (upward small, upward large, downward small, and downward large), with 12 rounds per type. By referring to the previously mentioned dispersion from the mean, the standard deviation of the AI forecasts was similar to that of the human forecasts (5.55 vs. 6.03). The exact demand distribution was unknown to the subjects, and we randomly selected data from products with an average demand of 10.

3.2. Participants

Using Amazon Mechanical Turk (MTurk), we recruited subjects based in the United States who had good reputations, i.e., at least 100 HITs completed with a 95% approval rating. We assigned them a qualification ID for each study to prevent retakes. A total of 68 subjects participated in our study. One subject answered our attention check incorrectly, and we excluded his results. The participants were balanced in terms of gender (55.2% male vs. 44.8% female), with an average age of 36.7 years, and most had a bachelor's degree (68.7%). Most of our subjects (76.1%) had good computer knowledge from school or work, and slightly more than half had forecasting knowledge (56.7%). On average, subjects earned a bonus of USD 2.20 (plus an extra USD 1 fixed participation fee), and each spent 17.2 min completing the study.

3.3. Variables

Table 2 shows the variables used in our analyses. A key dependent variable in Study 1 was Human_selected, which is a binary variable indicating whether the human forecast was selected. Using this variable, we aimed to understand when subjects were more likely to choose a human-adjusted forecast than an AI forecast when neutral guidance was provided without referencing the attributes of the adjustment made (e.g., direction and magnitude of adjustment). We used the set of independent variables[3] shown in Table 2 to understand the circumstances where a human-adjusted forecast was more likely to be selected.

3.4. Results

In the first study, we aimed to understand the natural behavior of the subjects in the presence of neutral guidance. Subjects chose the human forecasts in nearly half of the cases (48.4%), and thus they chose the AI forecasts in the other half. This behavior aligned with the instructions that they received, and it reflected an overall equal chance of AI and human superiority. Moreover, this result is consistent with the findings obtained by Castelo et al. (2019) and Dietvorst et al. (2018) who found no evidence for

[3] We collected some information about the subjects' backgrounds such as their prior knowledge of using computers and forecasting. However, we found no significant effects from their backgrounds on the dependent variables in our studies, i.e., the likelihood of accepting human forecasts and the likelihood of following the guidance.

Table 2
Overview of variables.

Variable	Description
Dependent variables	
Human_selected	Binary variable indicating whether Subject s in Round i selected human forecast HF_i over AI forecast AIF_i as the final demand forecast for Round i : FF_i
Independent variables	
Human closer to mean	Binary variable indicating whether human forecast HF_i was closer to the constant average demand (10 units) compared with the associated AI forecast AIF_i in Round i
|Adjustment| /AI forecast	Numeric variable showing the relative magnitude of adjustment Adj_i compared with AI forecast AIF_i in Round i
Earlier rounds	Binary variable indicating whether Round i belonged to the page of the first eight rounds
Later rounds	Binary variable indicating whether Round i belonged to the page of the last eight rounds
Directly after feedback	Binary variable indicating whether Round i was a round that came immediately after sharing feedback
Large magnitude	Binary variable indicating whether the magnitude of adjustment Adj_i was more than 5 units or less than 5 units (small adjustment) in Round i
Upward direction	Binary variable indicating whether adjustment Adj_i increased AI forecast AIF_i (upward adjustment) or decreased AIF_i (downward adjustment) in Round i

Table 3
Results of logistic regression analysis for Study 1 (dependent variable "Human_selected").

Variable		
Human closer to mean	0.8816***	(0.0789)
|Adjustment|/AIforecast	−0.0230	(0.0357)
Earlier rounds	0.0248	(0.1011)
Later round	0.1081	(0.1005)
Directly after feedback	−0.3749**	(0.1184)
Large magnitude	−0.0006	(0.0780)
Upward direction	0.3463***	(0.0794)
Constant	−0.5871***	(0.0859)

Standard errors are shown in parentheses; *p-value < 0.05, **p-value <0.01, ***p-value <0.001.

either algorithm aversion or appreciation when providing subjects with aggregated prediction performance.

We conducted logistic regression to analyze the factors that affected the selection of human forecasts (see Table 3). Unsurprisingly, the subjects tended to select forecasts closer to the average demand (p < 0.001, z = 11.170), which is consistent with the well-known mean-anchoring effect (Bloomfield & Kulp, 2013; Schweitzer & Cachon, 2000). In addition, the results obtained by logistic regression analysis indicated that the subjects significantly preferred an upward direction for adjustments to downward adjustments (p < 0.001, z = 4.360). Subjects were more likely to choose a human forecast, which resulted from adjusting an AI forecast in an upward direction. Moreover, subjects appeared to have a higher tendency to choose AI forecasts immediately after receiving feedback (coefficient value for directly after feedback of –0.3749 (p < 0.01, z = –3.165) as shown in Table 3). However, this effect did not persist when we investigated it for two rounds after receiving feedback (coefficient value of –0.0796, with p = 0.357, z = –0.921). Interestingly, we found no significant effect of the magnitude of adjustments, but large adjustments were assumed to make larger changes, and thus they were expected to capture more attention than small adjustments.

3.5. Discussion

Overall, the human forecasts were selected in nearly half of the cases, which agreed with the information provided that human forecasts improved the AI forecasts in half the cases. However, in this study, no information was provided to the subjects when the human forecast was really better, and this was left to the judgment of the subjects.

We found that the mean-anchoring effect and optimism bias influenced human choices. The results showed that subjects tended to select the forecast closer to the average demand, which is consistent with previous research into the mean-anchoring effect (Schweitzer & Cachon, 2000). The mean demand was the first piece of information offered to subjects when they were making decisions, so it acted as an anchor. Our results also indicate that subjects tended to choose a human forecast with an upward adjustment more often than one with a downward adjustment, and this tendency may have been related to optimism bias. Hence, our results showed that common human biases and heuristics were also persistent in environments where humans interacted with algorithms.

We also found that subjects were more likely to choose the AI forecast after receiving feedback, but this effect did not last long. We found no significant effect of the size of the adjustments on the forecast selection, thereby implying that the magnitude of the changes was less important when choosing between AI forecasts and human forecasts. Overall, we needed to retain the implications for improving the design and communication of AI forecasts, such as providing feedback, reducing the anchoring effect, and mitigating the optimism bias.

4. Study 2: Decomposed guidance with different levels of salience and detail

In Study 2, we provided decomposed guidance about the improvement probability of adjustments made by referring to their attributes, i.e., direction and magnitude. In the second study, we aimed to investigate the effects

Table 4
Experimental design for Study 2.

		Salience of information	
		Less salient 20% difference in improvement probabilities	More salient 40% difference in improvement probabilities
Detail	**Less detailed:** Adjustment direction	*Treatment 1 (T1)*	*Treatment 2 (T2)*
	More detailed: Adjustment direction and magnitude	*Treatment 3 (T3)*	*Treatment 4 (T4)*

Table 5
Improvement probabilities based on direction and magnitude for each treatment in Study 2.

			Less salient		More salient	
			Direction of adjustments			
			Downward	*Upward*	*Downward*	*Upward*
Less detailed			60%	40%	70%	30%
More detailed	Magnitude of adjustments	*Large*	60%	40%	70%	30%
		Small	40%	40%	30%	30%

of different guidance formats that varied in salience and level of detail as part of an effort to understand when guidance usage increased.

4.1. Experimental design

We designed a full-factorial (2×2) setting to evaluate guidance in four different treatments: improvement probabilities revealed with respect to only one attribute (direction) of adjustments in two forms of less and more salience, as well as improvement probabilities revealed respective to both attributes, direction and magnitude, again in two forms of less and more salience. Table 4 summarizes all four treatments. Using this design, we investigated the effect of the salience of guidance (referring to H1) and the interaction effect between salience and level of detail (referring to H2).

In a similar task using the same data employed in the first study, a subject had the role of a supervisor in a demand forecasting task. In addition to all the information provided in the first study, we provided improvement probabilities in the guidance based on the two attributes (direction and magnitude) with important roles in the quality of adjustments made by humans to AI forecasts. Probabilities in the less detailed conditions were provided based solely on one attribute: direction. In the more detailed conditions, the improvement probabilities were provided based on both direction and magnitude. Based on the effect of distance in numbers (Kaufmann et al., 2005), we influenced the salience of information by using the difference in improvement probabilities provided in the guidance. In the less salient condition, there was a 20% difference in the improvement probabilities between downward and upward adjustments. In the more salient condition, we increased this difference to 40%. In all conditions, the aggregated improvement probability of adjustments was 50%, as also applied in Study 1. The different aspect of this study is that these probabilities were decomposed according to different directions and magnitudes of adjustments, and the decomposed probabilities were provided together. Given this information, the subjects' main task was to choose either an AI forecast or a human-adjusted forecast. We kept the overall instructions identical to those in Study 1. Other aspects, such as the phases of the experiment and payoff procedures, were also the same as those applied in Study 1.

In this study, the guidance stated the improvement probabilities of adjustments at an aggregated level and also with respect to their attributes. Table 5 summarizes the improvement probabilities given to each attribute. The improvement probabilities of upward adjustments were characterized by two factors compared with downward adjustments in our design: the improvement possibilities were always lower, reflecting worse performance of upward adjustments in line with prior research; and they were identical across less detailed and more detailed conditions, which made the analyses more interesting. There should have been no significant effect on the usage of guidance between different conditions focused only on upward adjustments. Moreover, any significant effects should have been found by focusing only on downward adjustments.

The second study contained the same guidance statement as the first: "On average, the human forecast is better than the AI forecast in 50% of the cases, i.e., the AI forecast is also better in 50% of the cases", but it also stated that: "The probability that the human forecast is better than the AI forecast differs for [different attributes of adjustments]", to briefly introduce a table of probabilities that varied across treatments in terms of content, according to Table 5. To increase the visibility of this guidance to a level similar to that in Study 1, we used a different text color and placed it in a box. Fig. 4 shows a user screen for T1 in this study. Appendix D contains examples for other treatments.

4.2. Participants

Using MTurk, we recruited subjects based in the United States with good reputations, i.e., at least 100 HITs completed with a 95% success rate. We assigned them a

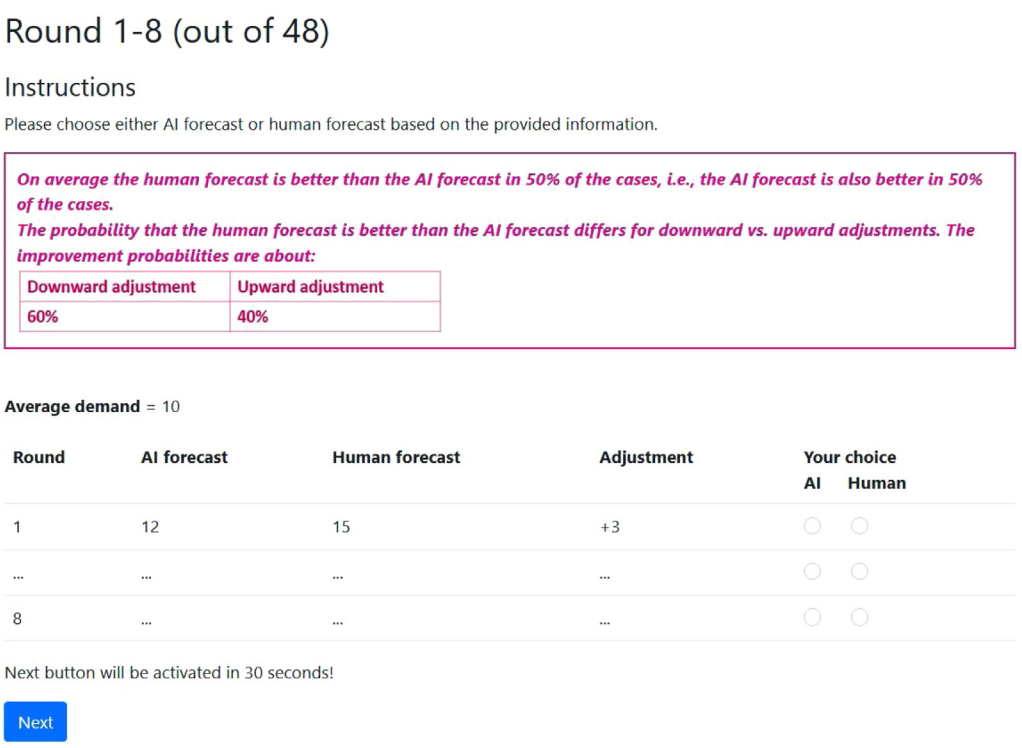


Fig. 4. User's screen for T1 in Study 2.

qualification ID in each study to prevent retakes. We recruited 267 subjects and 32 failed our attention checks. After excluding these individuals, we obtained data from 63 subjects for each of the less detailed conditions (T1 and T2), 56 subjects for the less salient but more detailed conditions (T3), and 53 subjects for the more salient with more detailed conditions (T4). The participants were balanced in terms of gender (55.9% male vs. 44.1% female), with an average age of 38.6 years, and most had a bachelor's degree (71.1%). Most of our subjects (86.0%) had good computer knowledge from school or work and forecasting knowledge (74.2%). On average, subjects earned a $2.2 bonus (plus an extra $1 fixed participation fee) and spent 19.3 min completing the study.

4.3. Variables

In addition to the variables applied in Study 1, we defined and used two new dependent variables: "Guidance_followed" and "Average_guidance_followed". The former is a binary variable used to indicate whether a subject made a decision in line with the guidance. We used "Guidance_followed" to specifically understand what motivated subjects to follow guidance at an individual level. For instance, if guidance implies better performance for downward adjustments compared with upward adjustments, this variable equals 1 when a subject selects a human forecast with a downward adjustment or chooses an AI forecast in the case of an upward adjustment. Suppose that the human forecast equals 8 and the AI forecast equals 15. In this case, the human forecast is 5 units smaller than the AI forecast, which means that there is a large downward adjustment to the AI forecast. When guidance reports a better improvement probability for a downward adjustment and the subject chooses the human-adjusted forecast, the dependent variable "Guidance_followed" is 1; otherwise, it is 0. In a case where guidance details the improvement probabilities regarding the *magnitude* of adjustments and reports a better improvement probability for a large downward adjustment, by the same choice, the dependent variable "Guidance_followed" is 1; otherwise, if the subject chooses the AI forecast over the human forecast, this variable is 0.

"Average_guidance_followed" refers to the frequency of decisions that the participants made in line with the guidance per round on average. In other words, this variable reveals how often the participants followed the guidance in each round on average. We used "Average_guidance_followed" to compare different treatments. For example, if a subject chooses all human forecasts with downward adjustments, "Average_guidance_followed" is 1, and this is the maximum value of this variable in an ideal case. If a subject chooses half of the human forecasts with downward adjustments, this variable is 0.5.

4.4. Results

4.4.1. Human forecast selections

Table 6 shows the percentages of decisions where human forecasts were selected across different treatments. In addition to the decomposed improvement probabilities according to adjustment direction and magnitude, we provided instructions that indicated the overall equal likelihood of AI and humans being superior. In line with

Table 6
Percentages of decisions where human forecasts were selected.

			Select human %
Study 1		Neutral guidance	48.4
Study 2	Less detailed	Less salient (T1)	47.8
		More salient (T2)	47.9
	More detailed	Less salient (T3)	47.6
		More salient (T4)	46.6

this instruction, subjects chose the human forecasts in 47.5% of the decisions, which is similar to the percentage of human forecasts selected in Study 1 (48.4%). Indeed, the AI forecast and human forecasts were not significantly different in terms of their overall contributions to accuracy, i.e., either could be better or worse, and the distinction was evident only by looking at the performance of different adjustment attributes. In this study, this was a key reason for designing guidance that reported the performance of adjustments according to their attributes.

We conducted logistic regressions for each treatment by considering "Human_selected" as a dependent variable. Table 7 summarizes the regression outputs. Similar to Study 1, we observed a mean-anchoring effect. The likelihood of selecting human forecasts increased when these forecasts were closer to the mean demand compared with AI forecasts. In addition, subjects tended to choose AI forecasts immediately after receiving feedback. In less detailed treatments (T1) and (T2) where guidance reported improvement probabilities with respect to the direction of adjustments, we found a significant negative effect for the upward direction ($p < 0.05$, z(T1) = –2.077, z(T2) = –2.195), i.e., the likelihood of human selections decreased for upward directions. In our first study, we found a significant positive effect for upward direction as evidence for optimism bias. In the second study, we found the opposite effect by providing the improvement probabilities by referring to Table 5. The previously shown bias (optimism bias) disappeared when the decomposed guidance was provided. Subjects choose human forecasts more systematically, and relied less on their bias and heuristics. This negative effect remained after additional details regarding the magnitude were provided in a less salient manner in (T3). Large magnitude also appeared to have a significant positive effect ($p < 0.05$, z = 2.119). In the more detailed and more salient treatment (T4), a significant negative interaction was found between the upward direction and large magnitude ($p < 0.01$, z = –3.199), thereby indicating a negative effect of the upward direction only when the adjustment size was large.

4.4.2. Average guidance followed

Next, we considered "Average_guidance_followed" as our dependent variable and tested our hypotheses. "Average_guidance_followed" measures the frequency of decisions that the participants made in line with the guidance per round on average. In total, our task had 48 rounds: 24 rounds with downward adjustments and 24 rounds with upward adjustments. Table 8 shows the average and standard errors for this variable under different treatments in the following order: *Overall* considering all rounds, *Upward* focusing on rounds with only upward adjustments, and *Downward* focusing on rounds with only downward adjustments. It should be noted that our guidance design showed that the improvement probabilities of upward adjustments were identical per guidance with a particular level of salience, i.e., always 40% for less salient conditions and always 30% for more salient conditions, regardless of the level of detail. Thus, given a salience condition, the additional information from the magnitude of adjustments (large vs. small) did not alter the improvement probabilities of upward adjustments. However, the improvement probabilities of downward adjustments were dependent on the magnitude of the adjustments, and thus the level of detail (see Table 5). Therefore, according to our guidance design, sharing details should only have affected the decisions associated with downward adjustments, and it should not have affected the decisions made for rounds with upward adjustments.

Effect of salience: There was no significant difference between the average number of cases of guidance followed in less salient versus more salient conditions in the form of less detailed guidance (Overall: 0.55 vs. 0.54; p-value = 0.839; Upward: 0.57 vs. 0.56; p-value = 0.901; Downward: 0.52 vs. 0.52; p-value = 0.920;[4]) as well as in the form of more detailed guidance (Overall: 0.57 vs. 0.57; p-value = 0.682; Upward: 0.59 vs. 0.53; p-value = 0.198; Downward: 0.55 vs. 0.61; p-value = 0.106). Hence, we rejected H1 and concluded that our less salient guidance was followed as frequently as our more salient guidance when the level of detail was fixed. Consequently, subjects perceived the difference between two sets of probabilities (60% vs. 40% and 70% vs. 30%) similarly, and there was not any significant difference in following or using the less salient guidance compared with the more salient guidance.

Effect of additional detail: As noted above, we expected to find that due to our guidance design, sharing details should only have affected the decisions associated with downward adjustments, with no effects on decisions made for rounds with upward adjustments. First, we found no significant difference between the average number of guidance-followed cases in less detailed versus more detailed conditions when probabilities in the guidance were less salient (Overall: 0.55 vs. 0.57; p-value = 0.132, Upward: 0.57 vs. 0.59; p-value = 0.421, Downward: 0.52 vs. 0.55; p-value = 0.214). However, we found a significant difference when probabilities in guidance were more salient. The average number of guidance-followed cases did not differ between less detailed versus more detailed conditions overall (0.54 vs. 0.57; p-value = 0.604) and for upward adjustments (0.56 vs. 0.53; p-value = 0.478), but we observed a significant difference for downward adjustments (0.52 vs. 0.61; p-value = 0.023), which implies that when guidance was more salient, providing more detailed guidance increased its usage. Therefore, we found support for H2.

[4] The p-values were obtained using the Wilcoxon rank sum test as the assumption of normality for parametric tests was violated in all cases according to the Shapiro — Wilk normality test. We obtained p-values using the same test unless the name of the statistical test is stated.

Table 7
Results obtained by logistic regression for Study 2 (dependent variable: "Human_selected").

	Less detailed		More detailed	
Variable	T1: Less salient	T2: More salient	T3: Less salient	T4: Moresalient
Human closer to mean	0.5593***	0.5074***	0.6396***	0.2245*
	(0.0863)	(0.0862)	(0.0920)	(0.0947)
|Adjustment|/AIforecast	−0.0143	0.0676	0.0077	0.0363
	(0.0374)	(0.0388)	(0.0398)	(0.0417)
Earlier rounds	0.0523	0.0723	0.0933	−0.0547
	(0.1035)	(0.1027)	(0.1109)	(0.1135)
Later round	−0.0324	−0.0031	0.0418	−0.0647
	(0.1025)	(0.1022)	(0.1095)	(0.1119)
Directly after feedback	−0.3103*	−0.1138	−0.3083*	−0.2851*
	(0.1220)	(0.1207)	(0.1309)	(0.1349)
Large magnitude	0.0840	0.0107	0.2438*	0.8374***
	(0.1078)	(0.1072)	(0.1150)	(0.1193)
Upward direction	−0.2190*	−0.2303*	−0.2822*	0.3761**
	(0.1054)	(0.1050)	(0.1125)	(0.1165)
Large magnitude × Upward direction	−0.1175	0.0162	−0.2513	−0.5811**
	(0.1668)	(0.1661)	(0.1781)	(0.1816)
Constant	−0.1938*	−0.2405**	−0.2853**	−0.6713***
	(0.0884)	(0.0883)	(0.0941)	(0.0994)
N	3024	3024	2688	2544

Standard errors are shown in parentheses; *p-value < 0.05, **p-value < 0.01, ***p-value < 0.001.

Table 8
Average guidance followed.

		Overall		Upward		Downward	
Less detailed	Less salient (T1)	0.54 (0.25)	0.55 (0.24)	0.56 (0.27)	0.57 (0.26)	**0.52 (0.26)**	0.52 (0.26)
	More salient (T2)		0.54 (0.26)		0.56 (0.27)		**0.52 (0.26)**
More detailed	Less salient (T3)	0.57 (0.17)	0.57 (0.15)	0.56 (0.29)	0.59 (0.27)	**0.58 (0.15)**	0.55 (0.12)
	More salient (T4)		0.57 (0.20)		0.53 (0.30)		**0.61 (0.17)**

Standard errors are shown in parentheses.

4.5. Discussion

In our experiments, we provided subjects with information about the improvement probabilities of different types of adjustments made by humans to AI forecasts, i.e., upward/downward and large/small. This design was used to explore how different types of information affected the selection of human and AI forecasts.

In Study 1, we conducted logistic regression to analyze how different types of information affected the choices of human and AI forecasts in a demand forecasting task. We found that human forecasts were chosen more often when they were closer to the average demand, which is consistent with the mean-anchoring effect. In the presence of additional guidance, we found that the choice of human forecasts decreased when upward adjustments were involved, which was the opposite of the optimism bias observed in the absence of specific guidance in Study 1. Hence, we suggest that providing information about the improvement probabilities of different types of adjustments can reduce optimism bias and help subjects make more informed choices. Nevertheless, there seemed to be a trade-off between revealing additional details and the performance of subjects because we observed a significant negative interaction effect for the attributes of adjustments on the choice of human forecasts, especially when adjustments were upward and large.

Moreover, in this study, we analyzed the effects of different types of guidance on the frequency of decisions that followed the guidance in a demand forecasting task. By considering "Average_guidance_followed" as a dependent variable to measure how often the subjects made decisions that matched the guidance in each round, we compared the average and standard error for this variable across different treatments that varied in terms of the level of detail and salience of the guidance.

In this study, we obtained two main findings. First, we found no significant difference between using more salient guidance and less salient guidance when the amount of information was held constant, i.e., in either less or more detailed treatments. Second, revealing more details in the guidance could increase its usage when the guidance was more salient. Moreover, subjects could consider the relative magnitudes of adjustments rather than only the absolute magnitude when making their decisions, but adding further details made this more difficult and statistically insignificant. In summary, we found greater use of more salient and more detailed guidance than other forms. Previous studies obtained mixed findings about revealing additional information, but we suggest providing additional information might be helpful under certain circumstances rather than merely believing in their effectiveness or ineffectiveness.

Table 9
Average MAE values for supervisors' decisions and benchmarks.

		MAE
Neutral guidance		4.98 (0.58)
Less detailed	Less salient (T1)	4.92 (0.64)
	More salient (T2)	4.88 (0.65)
More detailed	Less salient (T3)	4.83 (0.65)
	More salient (T4)	4.95 (0.61)
Human adjusted forecasts		5.21
AI forecasts		5.02
Fully following guidance in (T1) and (T2)		4.23
Fully following guidance in (T3) and (T4)		3.98

Standard errors are in parentheses.

4.6. Forecasting accuracy comparison

In further analysis, we compared the forecasting performance of subjects across the different decision guidance conditions examined in Studies 1 and 2. We used the mean absolute errors (MAE) of forecasts to indicate the forecasting accuracy. A lower MAE indicated higher forecasting accuracy. We also selected the following three benchmarks: (i) when human-adjusted forecasts were always selected (the case in the absence of a forecast supervisor); (ii) when AI forecasts were always selected; and (iii) when one chose forecasts by fully following each specific form of guidance to represent an ideal case.

Table 9 shows the average MAE values for the forecasts. First, we found no significant differences in performance across our study treatments. The characteristics of our setting, such as similar dispersion of the AI forecasts and human forecasts, did not allow for a significant decrease in the MAE. Overall, even with neutral guidance, subjects acting as forecast supervisors performed better than those in the cases where human-adjusted forecasts were always adopted and AI forecasts were always adopted. This implies that another decision process structure layer with a supervisor and appropriate guidance could increase the accuracy.

5. Conclusion

Judgmental adjustments are an indispensable part of forecasting, and they can involve more than one layer of human experts, e.g., those who modify or correct forecasts generated by a system and those who supervise human adjustments to system forecasts and finalize these forecasts. There are various methods for improving the forecast accuracy, including helping forecast supervisors make better choices, i.e., choosing human-adjusted forecasts that are more likely to be better than AI forecasts and reverting to AI forecasts when human adjustments are less likely to add value.

We conducted two experimental studies to determine the best methods to assist forecast supervisors. We asked subjects to take the role of a forecast supervisor and choose between an AI forecast and a human-adjusted AI forecast. We designed the decision guidance by providing the improvement probability for the adjustments. Using this guidance, subjects had the opportunity to improve by selecting the human-adjusted forecast over the original AI forecast.

In the first study, we provided the improvement probability for adjustments only at an aggregated level, without referencing the attribute of the adjustment, and ensured that the improvement probabilities indicated that the value of an adjustment (human adjusted forecast) equaled that of no adjustment (AI forecast). In this neutral setting, we found that subjects relied heavily on heuristics. They preferred to choose the forecast closer to the average demand by referring to mean anchoring. In addition, consistent with overoptimism bias, subjects were more likely to choose human forecasts with an upward adjustment. However, there was no tendency to prefer either human- or AI-generated forecasts.

In the second study, using a full-factorial design experiment, we tested four types of guidance that varied in terms of the level of detail and salience: less salient, less detailed (T1); more salient, less detailed (T2); less salient, more detailed (T3); and more salient, more detailed (T4). We varied the level of salience according to the difference in the improvement probabilities given in the guidance, i.e., the guidance was more salient when the difference was larger. Moreover, we provided improvement probabilities with respect to the direction of adjustment in less detailed treatments, whereas the improvement probabilities were given in terms of both the direction and magnitude of adjustment in more detailed treatments. We found that the likelihood of guidance being followed did not depend on the level of salience in guidance in less detailed conditions. However, we found that providing more details increased guidance use by subjects only when guidance was more salient (see Fig. 5).

There was no significant difference in the frequency of guidance being followed between different levels of detail given in the guidance, thereby demonstrating that additional details did not contribute to the use of guidance when guidance was less salient. Moreover, we found that sharing guidance with supervisors generally improved the forecasting performance by decreasing the MAE of the final forecasts compared with a situation where human-adjusted forecasts always served as final forecasts. The reduction in errors implies that given another process layer by human supervisors, sharing guidance can successfully avoid the finalization of adjustments with a greater potential of decreasing the forecast accuracy.

One key difference between our study and others (see Dietvorst et al., 2015 or Perera et al., 2022) is that we did not communicate a single forecast. Instead, we considered a setting where multiple humans in the process had to deal with different considerations, such as what the system suggested, what another human suggested based on that, and what was the best decision to finalize, i.e., we intended to guide decision-makers toward a better choice based on the conditions. At an aggregated level, we found evidence for a common cognitive bias: overoptimism (Eroglu & Croxton, 2010; Franses & Legerstee, 2009). Several studies found inferior performance for upward adjustments compared with downward adjustments, mainly when the adjustment size was large

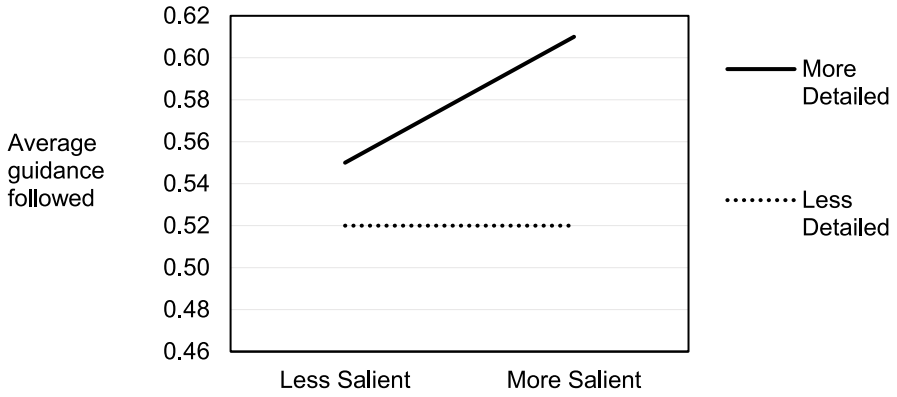


Fig. 5. Interaction between the salience and level of detail in guidance.

(Davydenko & Fildes, 2013; Fildes et al., 2009; Khosrowabadi et al., 2022). Therefore, we included the direction and magnitude of adjustments as key attributes in the guidance provided as probabilities of improvement. In accordance with previous research into decision guidance, especially the study by Bolton and Katok (2018), we organized various guidance options.

We tested the effects of communicating different improvement probabilities for adjustments with small and large discrepancies, which we referred to as less and more salient guidance, according to the numerical distance effect (Buckley & Gillman, 1974; Kaufmann et al., 2005; Moyer & Landauer, 1967; Pinel et al., 2004). We assumed that we would find better performance with more salient guidance. However, we found no significant effect on salience provided that we fixed the amount of information, which we referred to as the level of detail. Inspired by the study by You et al. (2022), we investigated the effect of elaboration on the performance of decision-makers. It is important to provide adequate information because an excessively small amount of information may require additional searching and overthinking, but an excessively large amount of information can be misleading and result in cognitive overload (Hunter, 2004; Jacoby et al., 1974; Kahneman, 2003). Previous studies, including those by Castelo et al. (2019) and Dietvorst et al. (2018), found no significant preference for a particular source of advice provided that the performance was reported at an aggregated level. However, You et al. (2022) found that performance elaboration diminished the tendency to follow system advice. Based on Corman's (1957) suggestion that more information with less explicitness may be as effective as less information with more explicitness, we varied the amount of information by reporting the improvement probabilities based on either the direction of adjustments alone or both the direction and magnitude of adjustments. We found evidence to support Corman's (1957) suggestion, because we observed that providing more information increased the effectiveness of the guidance when the information was more salient.

Our findings have practical implications for designers of forecast support systems. First, sharing some immediate, simple feedback is helpful, but sharing some forms of guidance to communicate the performance of different forecasts also helps to achieve improvements. Second, cognitive biases do exist and we should assess how they might be harmful. Providing excessive or irrelevant information can trigger cognitive biases (Tversky & Kahneman, 1974), and thus unnecessary information should not be highlighted to avoid exacerbating these biases. A meta-analysis by Kluger and DeNisi (1996) demonstrated that poorly understood task characteristics can indeed lead to the adverse effects of feedback intervention on performance. Sleiman, Sigurjonsdottir, Elnes, Gage, and Gravina (2020) showed that the effect of feedback was not simple and could be moderated by many factors, such as feedback specificity, frequency, and the individual's feedback orientation. Therefore, we need to think carefully about the type and amount of feedback provided to avoid reinforcing negative behaviors. Third, clear and noticeable information is key. If there is a tendency to elaborate on performance, designers should ensure that adequate attention is directed toward key information. For example, suppose that the aim is to direct individuals toward a specific action from a range of options. In this case, the success rate of that action should be significantly higher than the success rate of the other actions to ensure that the guidance is highly likely to be effective even with additional information.

Similar to all experimental studies, we had to make choices that should be discussed. We created and evaluated different types of guidance, which varied in terms of their noticeability and detail, and to the greatest possible extent that our real-world data allowed. However, to discover the full potential of our proposed guidance in practice, further studies are required using different probabilities and by considering different real-world data to validate our findings. To avoid falsifications, we did not manipulate the success probabilities, which were the main elements of our guidance, because the probabilities represented by guidance should reasonably match reality. One can also consider studying the effectiveness of guidance by simulating data to contain controllability, although the simulated data will not capture all real-world data characteristics. There are many other interesting avenues for future studies. First, manipulating the time available to decision-makers can impact the quality of guidance and final decisions, such as in fast- and slow-thinking systems. Moreover, replicating the experiments

using subjects with various levels of contextual knowledge or from different countries could help to investigate the impacts of cultural norms and background knowledge on their decisions.

Moreover, we encourage replication of our study with other subject pools, and ideally with field experts, because MTurkers may not represent their specific behaviors. For instance, experts who have worked with traditional forecasting models for years may exhibit specific biases or preferences toward AI forecasting, which is often viewed as a black box. They might consider variables for adjustments that are typically accounted for by AI, thereby leading to a double-accounting problem. In addition, they may compare AI forecasts with their own estimates based on their expertise, potentially leading to confirmation bias. Exploring scenarios where subjects make a final forecast that blends AI and human forecasts, rather than choosing one over the other, could also be a fruitful direction for future research.

CRediT authorship contribution statement

Naghmeh Khosrowabadi: Data curation, Formal analysis, Investigation, Methodology, Validation, Writing – original draft, Conceptualization, Writing – review & editing. **Kai Hoberg:** Conceptualization, Data curation, Investigation, Supervision, Writing – original draft, Writing – review & editing. **Yun Shin Lee:** Formal analysis, Investigation, Methodology, Supervision, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data and code availability

All data and code are openly shared within a reproducibility package available on GitHub at the following link: https://github.com/kaihoberg/Khosrowabadi-et-al.-2024---IJF---Guiding-Supervisors-in-AI-enabled-Forecasting.

Appendix. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.ijforecast.2024.08.001.

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